ER Patient Experience Dashboard

Data Inputs, KPIs & Calculation Logic (Code-Based)

Purpose

This dashboard monitors **patient experience drivers in the Emergency Room** by combining:

- Operational timing metrics
- Communication effectiveness
- Behavioral outcomes (LWBS, revisits)
- Satisfaction and risk indicators

All values shown are **explicit mock inputs defined in the frontend code** to validate KPI meaning, relationships, and visualization behavior.

1. Stat Cards (Top-Level KPIs)

These are **headline indicators** summarizing ER performance.

1.1 Average Time to First Clinical Contact

Value (code):

```
"12 min"
```

Meaning:
Average time from arrival to first clinical interaction.

Intended calculation:

```
AVG(first_contact_time - arrival_time)
```

1.2 % Receiving Timely Communication

Value (code):

```
"78%"
```

Meaning:
Patients who received at least one communication within a defined time threshold.

Intended calculation:

```
(visits_with_communication_within_threshold / total_visits) × 100
```

1.3 Overall Patient Satisfaction

Value (code):

```
"4.2 / 5"
```

Meaning:
Average patient satisfaction score.

Intended calculation:

```
AVG(satisfaction_score)
```

1.4 LWBS Rate

Value (code):

```
"5%"
```

Meaning:
Patients who left without being seen.

Intended calculation:

```
(LWBS_visits / total_visits) × 100
```

1.5 Revisit Rate (72 Hours)

Value (code):

```
"8%"
```

Meaning:
Patients returning within 72 hours of discharge.

Intended calculation:

```
(visits_with_revisit_within_72h / total_discharges) × 100
```

1.6 High Dissatisfaction Risk Visits

Value (code):

```
"10%"
```

Meaning:
Visits flagged by the dissatisfaction risk model.

Intended calculation:

```
(high_risk_visits / total_visits) × 100
```

2. Charts & KPIs (Exact Inputs + Logic)

2.1 Arrival → First Contact Trend

Component: `ArrivalToFirstContactTrend`

Inputs (code):

```
data={[
  { date: "2024-01-01", avgMinutes: 18 },
  { date: "2024-01-02", avgMinutes: 21 },
  { date: "2024-01-03", avgMinutes: 24 },
  { date: "2024-01-04", avgMinutes: 20 },
  { date: "2024-01-05", avgMinutes: 22 },
]}
granularity="daily"
threshold={20}
```

Logic:
Tracks daily average time to first contact and flags SLA breaches above 20 minutes.

2.2 Waiting Time by Triage Level

Component: `WaitingTimeByTriage`

Inputs (code):

```
data={[
  { level: "Level 1", minutes: [5, 7, 8, 6, 10] },
  { level: "Level 2", minutes: [10, 12, 15, 11, 14] },
  { level: "Level 3", minutes: [20, 25, 18, 22, 30] },
  { level: "Level 4", minutes: [30, 35, 28, 32, 40] },
  { level: "Level 5", minutes: [40, 45, 42, 38, 50] },
]}
threshold={30}
```

Logic:
Shows waiting-time distributions per triage level and highlights overload beyond 30 minutes.

2.3 Communication Coverage by Section

Component: `CommunicationCoverageBar`

Inputs (code):

```
sections={["Triage", "ER Room A", "ER Room B", "Observation", "ICU"]}
coverage={[95, 80, 85, 70, 60]}
threshold={85}
```

Logic:
Percentage of visits with documented communication per ER section.

2.4 Time to First Communication Trend

Component: `TimeToFirstCommunicationLine`

Inputs (code):

```
dates={["2025-10-01", "2025-10-02", "2025-10-03", "2025-10-04"]}
avgMinutes={[12, 15, 10, 18]}
threshold={15}
```

Logic:
Tracks responsiveness of communication after arrival.

2.5 Length of Stay vs Satisfaction

Component: `LOSvsSatisfaction`

Inputs (code):

```
lengthsOfStay={[30, 45, 60, 75, 90, 120]}
satisfactionScores={[95, 90, 85, 70, 60, 50]}
threshold={70}
```

Logic:
Visualizes satisfaction decay as length of stay increases.

2.6 First Contact Delay Impact on Satisfaction

Component: `FirstContactDelayLine`

Inputs (code):

```
buckets={["0-5 min", "5-10 min", "10-15 min", "15-30 min", "30+ min"]}
avgSatisfaction={[95, 88, 80, 70, 60]}
threshold={10}
```

Logic:
Demonstrates how early delays disproportionately affect satisfaction.

2.7 LWBS Rate by Section

Component: `LWBSRateBar`

Inputs (code):

```
sections={["Triage", "Fast Track", "Main ER", "Resuscitation"]}
lwbsRates={[3.2, 5.8, 9.4, 2.1]}
threshold={5}
```

Logic:
Identifies where patients are most likely to leave before being seen.

2.8 Revisit Rate vs Communication

Component: `RevisitVsCommunicationBar`

Inputs (code):

```
categories={["Communication Provided", "No Communication"]}
revisitRates={[8.5, 18.2]}
threshold={12}
```

Logic:
Quantifies the impact of communication on short-term returns.

2.9 Satisfaction by Shift & Pressure Level

Component: `SatisfactionByShiftPressure`

Inputs (code):

```
data={[
  { shift: "Morning", pressureLevel: "Low", satisfaction: 88 },
  { shift: "Morning", pressureLevel: "Medium", satisfaction: 75 },
  { shift: "Morning", pressureLevel: "High", satisfaction: 62 },
  { shift: "Afternoon", pressureLevel: "Low", satisfaction: 85 },
  { shift: "Afternoon", pressureLevel: "Medium", satisfaction: 70 },
  { shift: "Afternoon", pressureLevel: "High", satisfaction: 55 },
  { shift: "Night", pressureLevel: "Low", satisfaction: 80 },
  { shift: "Night", pressureLevel: "Medium", satisfaction: 68 },
  { shift: "Night", pressureLevel: "High", satisfaction: 50 },
]}
```

Logic:
Identifies operational conditions where patient experience is most fragile.

2.10 Dissatisfaction Risk by ER Section

Component: `DissatisfactionRiskBySection`

Inputs (code):

```
data={[
  { section: "Triage", riskScore: 85 },
  { section: "Pediatrics", riskScore: 70 },
  { section: "Trauma", riskScore: 55 },
  { section: "Cardiology", riskScore: 40 },
  { section: "General", riskScore: 25 },
]}
```

Logic:

Aggregated output of a dissatisfaction risk model by ER section.